Chapter 12. Classification of Surfaces

Fundamental groups of surfaces, homology of surfaces, cutting and pasting, the classification theorem

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12.1 Fundamental groups of surfaces

A **polygonal region** P in the plane is the convex hull of points $p_0, \dots, p_{n-1}$ ($n \geq 3$) lying on a circle in counterclockwise order; its **edges** are the segments $p_{i-1}p_i$ ($p_n = p_0$). An **orientation** of an edge is a choice of initial point; a **labelling** assigns a label to each edge. If edges with the same label are pasted by the positive linear maps respecting orientations, we obtain a quotient space X. A labelling and orientation are recorded by a **labelling scheme** $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$: the k-th edge has label a_{i_k} and is oriented counterclockwise if $\epsilon_k = +1$, clockwise if $\epsilon_k = -1$. Several disjoint polygonal regions with one scheme each are handled the same way. For example, $aba^{-1}b^{-1}$ gives the torus, $abab$ the projective plane (the 2-fold dunce cap), and $aa^{-1}bb^{-1}$ the sphere.

Theorem 12.1 (74.1)

The space X obtained from finitely many polygonal regions by pasting edges according to a labelling scheme is compact Hausdorff.

Proof. Compactness is clear. By Lemma 11.34 (the regions are normal) it suffices that the quotient map π is closed, i.e. $\pi^{-1}\pi(C)$ is closed for closed C. For an edge e, the set $\pi^{-1}\pi(C) \cap e$ is the union of $C \cap e$ and the images $h_i(C \cap e_i)$ under the pasting homeomorphisms $h_i : e_i \to e$ of the edges e_i pasted to e; this is compact, hence closed. And $\pi^{-1}\pi(C)$ is the union of C and these finitely many sets. ∎

Theorem 12.2 (74.2)

Let X be obtained from a polygonal region P by a scheme $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$, with quotient map π. If π maps all vertices to one point x_0 and $a_1, \dots, a_k$ are the distinct labels, then $\pi_1(X, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_k$ by the least normal subgroup containing $(\alpha_{i_1})^{\epsilon_1} \cdots (\alpha_{i_n})^{\epsilon_n}$.

Proof. $A = \pi(\operatorname{Bd} P)$ is a wedge of k circles, one for each label (each edge image is an arc whose end points are identified to x_0). For each i choose an edge labelled a_i, let f_i be the linear path along it in the counterclockwise direction and $g_i = \pi \circ f_i$; by Theorem 11.27, the $[g_i]$ (or their inverses) are free generators of $\pi_1(A, x_0)$. $P \cong B^2$, and π maps $\operatorname{Int} P$ bijectively onto $X - A$. The loop f going once counterclockwise around $\operatorname{Bd} P$ generates $\pi_1(\operatorname{Bd} P)$, and $\pi \circ f$ is the product $g_{i_1}^{\epsilon_1} * \cdots * g_{i_n}^{\epsilon_n}$, where g^{-1} means $\bar{g}$ (the edge labelled a_{i_j} with $\epsilon_j = -1$ is traversed against its orientation). Theorem 11.31 gives the result. ∎

The n**-fold torus** $T_n = T\# \cdots \#T$ is obtained from a $4n$-gon by the scheme $(a_1b_1a_1^{-1}b_1^{-1}) \cdots (a_nb_na_n^{-1}b_n^{-1})$; for $n = 2$ it is the double torus of Theorem 9.46. The m**-fold projective plane** P_m is obtained from a $2m$-gon by $(a_1a_1) \cdots (a_ma_m)$ ($m > 1$); set $P_1 = P^2$.

Theorem 12.3 (74.3)

$\pi_1(T_n, x_0)$ is the quotient of the free group on $\alpha_1, \beta_1, \dots, \alpha_n, \beta_n$ by the least normal subgroup containing $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, where $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. By Theorem 12.2 it suffices that all vertices are identified. In a block $aba^{-1}b^{-1}$ on vertices $v_0, \dots, v_4$: the edges labelled a paste v_0 to v_3 and v_1 to v_2; those labelled b paste v_1 to v_4 and v_2 to v_3. So $v_0, \dots, v_4$ are identified, and consecutive blocks share a vertex. ∎

Theorem 12.4 (74.4)

$\pi_1(P_m, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_m$ by the least normal subgroup containing $\alpha_1^2\alpha_2^2 \cdots \alpha_m^2$.

Proof. In a block aa on v_0, v_1, v_2, the two edges paste v_0 to v_1 and v_1 to v_2; consecutive blocks share a vertex. Apply Theorem 12.2. ■

12.2 Homology of surfaces

For path-connected X let $\pi_1 = \pi_1(X, x_0)$. If α, β are paths from x_0 to x_1, then $\hat{\alpha}$ and $\hat{\beta}$ induce the same isomorphism $\pi_1(X, x_0)/[\pi_1, \pi_1] \to \pi_1(X, x_1)/[\pi_1, \pi_1]$: with $g = \alpha * \bar{\beta}$, $\hat{g}([f]) = [g]^{-1}[f][g]$, which has the same coset as $[f]$ in the abelian quotient. This abelian group, determined up to canonical isomorphism, is the **first homology group** $H_1(X)$. A homeomorphism induces an isomorphism of fundamental groups, hence of H_1.

Theorem 12.5 (75.1)

Let $N \lhd F$ and $q : F \to F/N$, $p : F \to F/[F, F]$ the projections. Then p induces an isomorphism $\phi : q(F)/[q(F), q(F)] \to p(F)/p(N)$.

Proof. Let $r : p(F) \to p(F)/p(N)$ and $s : q(F) \to q(F)/[q(F), q(F)]$ be projections. $r \circ p$ kills N, so it induces $u : q(F) \to p(F)/p(N)$, and u kills commutators (the target is abelian), inducing ϕ. Likewise $s \circ q$ maps into an abelian group, so induces $v : p(F) \to q(F)/[q(F), q(F)]$, and $v \circ p = s \circ q$ kills N, so v kills $p(N)$ and induces ψ. Explicitly, $\phi(sq(x)) = rp(x)$ and $\psi(rp(x)) = sq(x)$; they are inverse to each other. ■

Corollary 12.6 (75.2)

Let F be free on $\alpha_1, \ldots, \alpha_n$, N the least normal subgroup containing x, and $G = F/N$. Then $G/[G, G]$ is the quotient of the free abelian group $F/[F, F]$ (basis $p(\alpha_i)$) by the subgroup generated by $p(x)$.

Proof. N is generated by the conjugates of x (Lemma 11.17), and all of them have image $p(x)$ in the abelian group, so $p(N)$ is generated by $p(x)$. Apply Theorem 12.5 and Theorem 11.21. ■

Theorem 12.7 (75.3)

$H_1(T_n)$ is free abelian of rank $2n$.

Proof. By Theorem 12.3 and Corollary 12.6, $H_1(T_n)$ is the free abelian group on α_i, β_i modulo the image of $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, which is 0 in an abelian group. ■

Theorem 12.8 (75.4)

The torsion subgroup $T(P_m)$ of $H_1(P_m)$ has order 2, and $H_1(P_m)/T(P_m)$ is free abelian of rank $m - 1$.

Proof. $H_1(P_m)$ is the free abelian group on $\alpha_1, \ldots, \alpha_m$ modulo the subgroup generated by $2(\alpha_1 + \cdots + \alpha_m)$. With $\gamma = \alpha_1 + \cdots + \alpha_m$, the elements $\alpha_1, \ldots, \alpha_{m-1}, \gamma$ form another basis, so $H_1(P_m) \cong \mathbb{Z}^{m-1} \times \mathbb{Z}/2$. ■

Theorem 12.9 (75.5)

The surfaces S^2, $T_1, T_2, \ldots$, $P_1, P_2, \ldots$ are topologically distinct.

Proof. Their first homology groups are 0, $\mathbb{Z}^{2n}$ and $\mathbb{Z}^{m-1} \times \mathbb{Z}/2$, pairwise non-isomorphic (by rank, Theorem 11.8, and by torsion). ■

12.3 Cutting and pasting

Let P have vertices $p_0, \dots, p_{n-1}$ and $1 < k < n-1$. Cutting P along p_0p_k gives Q_1 (vertices $p_0, \dots, p_k$) and Q_2 (vertices $p_k, \dots, p_n = p_0$); after translating Q_1 to be disjoint from Q_2, P is homeomorphic to the quotient of $Q_1 \sqcup Q_2$ pasting the two copies of p_0p_k linearly. Conversely, two disjoint regions pasted along one edge each can be represented by one region: move the vertices of one onto the circle of the other.

Theorem 12.10 (76.1)

Let X be obtained from m regions by the scheme $y_0y_1, w_2, \dots, w_m$, and let c be a label not appearing in it. If y_0 and y_1 have length at least 2, then X is also obtained from $m+1$ regions by the scheme $y_0c^{-1}, cy_1, w_2, \dots, w_m$. Conversely, the second scheme gives the same space as the first.

Proof. Cut the first region along the diagonal from p_0 to p_k, k the length of y_0, and label both copies of the diagonal c: it runs counterclockwise in the second piece and clockwise in the first. A composite of quotient maps is a quotient map, so pasting the two copies of c first and then the other edges gives the same space as pasting everything at once. ■

The following **elementary operations** on a scheme do not change the space. Two schemes are **equivalent** if one is obtained from the other by such operations; each operation has an inverse operation. (i) *Cut*: replace y_0y_1 by y_0c^{-1} and cy_1 (c new, y_0, y_1 of length ≥ 2). (ii) *Paste*: replace y_0c^{-1} and cy_1 by y_0y_1 (c appearing nowhere else). (iii) *Relabel*: replace all occurrences of a label by a new label, or change the sign of all exponents of one label. (iv) *Permute*: replace y_0y_1 by y_1y_0 (renumber the vertices). (v) *Flip*: replace $(a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$ by $(a_{i_n})^{-\epsilon_n} \cdots (a_{i_1})^{-\epsilon_1}$ (reflect the region). (vi) *Cancel*: replace $y_0aa^{-1}y_1$ by y_0y_1 if a appears nowhere else and y_0, y_1 have length ≥ 2. Indeed, cut to get y_0ab and $b^{-1}a^{-1}y_1$; in each region the two edges a, b can be merged into one edge c (the composite pasting is linear on each), giving y_0c and $c^{-1}y_1$; paste. (vii) *Uncancel*: the reverse of (vi).

Example 12.1

The Klein bottle $aba^{-1}b$ is P_2: cut to abc^{-1} and $ca^{-1}b$; permute the first to $bc^{-1}a$; flip the second to $b^{-1}ac^{-1}$ and permute it to $ac^{-1}b^{-1}$; paste along b to get $ac^{-1}c^{-1}a$; permute and relabel to $aabb$.

12.4 The classification theorem

A scheme is **proper** if each label appears exactly twice; elementary operations preserve properness. A proper scheme w for one region is of **torus type** if each label appears once with each exponent, and of **projective type** otherwise. In what follows, brackets only group letters, and y^{-1} is the formal inverse of a word y.

Lemma 12.11 (77.1)

If $w = [y_0]a[y_1]a[y_2]$ is proper (some y_i possibly empty), then $w \sim aa[y_0y_1^{-1}y_2]$.

Proof. Step 1: y_0 empty, $w = ay_1ay_2$. If y_1 is empty there is nothing to do. If y_2 is empty, flip to $a^{-1}y_1^{-1}a^{-1}$, permute and relabel to aay_1^{-1}. Otherwise cut into ay_1c^{-1} and cay_2; flip the first to $cy_1^{-1}a^{-1}$ and permute the second to ay_2c; paste along a to get $cy_1^{-1}y_2c$; permute and relabel to $aay_1^{-1}y_2$. Step 2: y_0 nonempty. If y_1, y_2 are empty, permute. Otherwise cut w into y_0ac^{-1} and cy_1ay_2; flip the first and permute to $y_0^{-1}ca^{-1}$; permute the second to ay_2cy_1; paste along a:

$y_0^{-1}cy_2cy_1$. Permute to $cy_2cy_1y_0^{-1}$; by Step 1 this is equivalent to $ccy_2^{-1}y_1y_0^{-1}$; flip to $y_0y_1^{-1}y_2c^{-1}c^{-1}$; permute and relabel to $aay_0y_1^{-1}y_2$. ■

Corollary 12.12 (77.2)

A scheme of projective type is equivalent to one of the same length of the form $(a_1a_1)(a_2a_2)\cdots(a_ka_k)w_1$ with $k \geq 1$ and w_1 empty or of torus type.

Proof. Some label occurs twice with the same exponent, which we may take to be $+1$; Lemma 12.11 gives $w \sim aaw_1$. If w_1 is not of torus type, write $w = aaz_0bz_1bz_2$ and apply the lemma with $y_0 = aaz_0$: $w \sim bbaaz_0z_1^{-1}z_2$. Each step lengthens the initial block of pairs, so the process ends with w_1 empty or of torus type. ■

In the next proof we use one more consequence of the operations.

Step E. If a does not occur in B, D_1, D_2, then the cyclic scheme $aBa^{-1}D_1D_2$ is equivalent to $aBa^{-1}D_2D_1$. We may assume D_1, D_2 nonempty. Permute to $Ba^{-1}D_1D_2a$, cut into $Ba^{-1}D_1c^{-1}$ and cD_2a, permute these to $D_1c^{-1}Ba^{-1}$ and acD_2, paste along a: $D_1c^{-1}BcD_2$. Permuting and relabelling $(c \to a^{-1})$ gives $aBa^{-1}D_2D_1$. Writing $aBa^{-1}D$ cyclically as $a^{-1}DaB$ and relabelling a, the same applies to the block B: *either block between a and a^{-1} may be cyclically rotated.*

Lemma 12.13 (77.3)

Let $w = w_0w_1$ be proper with w_1 of torus type containing no two adjacent terms with the same label. Then $w \sim w_0w_2$ with w_2 of the same length as w_1 and of the form $aba^{-1}b^{-1}w_3$, w_3 empty or of torus type.

Proof. Among the labels of w_1 let a be one whose two occurrences are closest; they are not adjacent, and we may assume a comes first. Let b occur between them, with exponent $+1$ after relabelling. Its other occurrence b^{-1} is not between a and a^{-1}. If it follows a^{-1}, then $w_1 = y_1aPbQa^{-1}Rb^{-1}y_5$; if it precedes a, exchange the roles of a and b (after changing exponents) to reach the same form. Cyclically, $w = aPbQa^{-1}Rb^{-1}S$ with $S = y_5w_0y_1$. Using Step E repeatedly: rotate the block PbQ between a and a^{-1} to bQP, obtaining $ab(QP)a^{-1}Rb^{-1}S$; rotate the block $(QP)a^{-1}R$ between b and b^{-1} to $a^{-1}RQP$, obtaining $aba^{-1}(RQP)b^{-1}S$; rotate the block $(RQP)b^{-1}S$ following a^{-1} to $b^{-1}SRQP$, obtaining $aba^{-1}b^{-1}T$ with $T = SRQP$. Finally, for any splitting $T = T_1T_2$, $aba^{-1}b^{-1}T_1T_2 \sim aba^{-1}b^{-1}T_2T_1$: rotate the block T_1T_2a between b^{-1} and b to T_2aT_1 and repeat the three rotations. With $T_1 = y_5w_0$ and $T_2 = y_1RQP$ we get, after permuting, $w_0aba^{-1}b^{-1}w_3$ where $w_3 = y_1RQPy_5$ consists of the remaining letters of w_1 with unchanged exponents, so it is empty or of torus type. ■

Lemma 12.14 (77.4)

If $w = w_0(cc)(aba^{-1}b^{-1})w_1$ is proper, then $w \sim w_0aabbccw_1$.

Proof. Permute to $cc[ab][ba]^{-1}[w_1w_0]$. By Lemma 12.11 read backwards (with $y_0 = ab$, $y_1 = ba$, $y_2 = w_1w_0$), this is equivalent to $[ab]c[ba]c[w_1w_0] = [a]b[c]b[acw_1w_0]$, which by the lemma is equivalent to $bb[ac^{-1}acw_1w_0] = [bb]a[c^{-1}]a[cw_1w_0]$, and again to $aa[bbccw_1w_0]$. Permute. ■

Theorem 12.15 (77.5, classification theorem)

A space obtained from a polygonal region by pasting its edges together in pairs is homeomorphic to S^2, to T_n, or to P_m.

Proof. The scheme w is proper of length at least 4. We show w is equivalent to one of (1) $aa^{-1}bb^{-1}$ (S^2), (2) $abab$ (P^2), (3) $(a_1a_1)\cdots(a_ma_m)$, $m \geq 2$ (P_m), (4) $(a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$ (T_n). If

w has length greater than 4 and contains adjacent terms aa^{-1}, cancelling (after permuting so that both remaining parts have length ≥ 2) gives a shorter scheme of the same type.

Step 1 (torus type). A torus-type scheme of length 4 is, up to permuting and relabelling, $aa^{-1}bb^{-1}$ or $aba^{-1}b^{-1}$. For longer w we use induction; we may assume no cancellation is possible. Lemma 12.13 gives $w \sim aba^{-1}b^{-1}w_3$ with w_3 nonempty of torus type, and again with no adjacent pair; applying the lemma with $w_0 = aba^{-1}b^{-1}$, and so on, we reach type (4).

Step 2 (projective type). For length 4, Corollary 12.12 gives $aabb$ (type 3) or aay with $y = b^{\pm 1}b^{\mp 1}$; writing $y = y_1^{-1}y_2$ with single letters, Lemma 12.11 gives ay_1ay_2, which relabels to $abab$. For longer w, Corollary 12.12 gives $(a_1a_1)\cdots(a_ka_k)w_1$. If w_1 is empty we have type (3). If w_1 has an adjacent pair, cancel and use induction. Otherwise Lemma 12.13 gives $(a_1a_1)\cdots(a_ka_k)aba^{-1}b^{-1}w_2$, and Lemma 12.14 turns this into $(a_1a_1)\cdots(a_{k-1}a_{k-1})aabba_ka_kw_2$, with a shorter torus part. Repeat. ■

12.5 Constructing compact surfaces

Let X be compact Hausdorff. A **curved triangle** in X is a subspace A with a homeomorphism $h : T \to A$ from a closed triangular region; edges and vertices of A are images of those of T. A **triangulation** of X is a finite collection of curved triangles covering X such that two distinct ones meet in nothing, a common vertex, or a common edge, and in the last case $h_j^{-1} \circ h_i$ is linear on the corresponding edge of T_i. Every compact surface is triangulable (Radó); we assume this.

Theorem 12.16 (78.1)

A compact triangulable surface is homeomorphic to the space obtained from finitely many disjoint triangular regions by pasting their edges together in pairs.

Proof. Let $A_1, \ldots, A_n$ be a triangulation with $h_i : T_i \to A_i$, T_i disjoint. Then $h : E = \bigsqcup T_i \to X$ is a closed continuous surjection, hence a quotient map. It suffices to show (a) each edge of each A_i is an edge of exactly one other triangle, and (b) if $A_i \cap A_j$ is a vertex v, there is a chain of triangles with vertex v from A_i to A_j, consecutive ones sharing an edge. Then h identifies exactly the points identified by the edge pastings.

Step 1 (b). Call two triangles with vertex v equivalent if joined by such a chain. If there were two classes, let B be the union of one and C of the rest; $B \cap C = \{v\}$. For any neighbourhood W of v missing the triangles not containing v, $W - v$ is the union of the disjoint relatively closed nonempty sets $(W \cap B) - v$ and $(W \cap C) - v$, so it is not connected. But v has a neighbourhood homeomorphic to an open disc and hence arbitrarily small neighbourhoods W with $W - v$ connected.

Step 2 (at least one other triangle). Let x be interior to an edge of a triangular region T. Its ϵ-neighbourhoods W in T have $W - x$ star-shaped about a point interior to T, hence simply connected. On the other hand, if x had a neighbourhood U homeomorphic to the open unit disc B with $x \mapsto 0$, then for every neighbourhood $V \subset U$ of x, choosing $B_\epsilon \subset V$, the inclusion $B_\epsilon - 0 \to B - 0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so it induces an isomorphism and $\pi_1(V - 0) \to \pi_1(B - 0) \cong \mathbb{Z}$ is surjective: $V - 0$ is not simply connected. Contradiction. Now if an edge e of A_i lay in no other triangle, a point x interior to e would have a neighbourhood in X contained in A_i, and a disc neighbourhood of x in X would give one in T_i.

Step 3 (at most one). Let $Y \subset \mathbb{R}^3$ be the union of $k \geq 3$ triangles meeting pairwise in a common edge e, and $x = 0$ interior to e. The union A of the edges other than e consists of k arcs with common end points; it retracts onto the union of three of them (map each extra arc homeomorphically onto one of the three, fixing end points), a theta space with nonabelian fundamental group (Example 11.1), so $\pi_1(A)$ is nonabelian. A is a deformation retract of $Y - 0$ (radial projection from 0 in each triangle). For any neighbourhood W of 0 in Y, choose ϵ with $Y' = \epsilon Y \subset W$; the inclusion $Y' - 0 \to Y - 0$ is

homotopic to $y \mapsto y/\epsilon$, a homeomorphism, so $\pi_1(W-0) \to \pi_1(Y-0)$ is surjective and $\pi_1(W-0)$ is nonabelian (at a base point in Y'). A disc neighbourhood W of 0 would have $\pi_1(W-0) \cong \mathbb{Z}$. If e were an edge of three or more triangles of the triangulation, a point interior to e would have a neighbourhood in X contained in their union, which is homeomorphic to such a Y. ■

Theorem 12.17 (78.2)

A compact connected triangulable surface is homeomorphic to a space obtained from one polygonal region by pasting its edges together in pairs.

Proof. By Theorem 12.16, X comes from triangular regions by a proper scheme. Whenever two different regions have edges with the same label, flip one if necessary and paste them along those edges (operation (ii)), reducing the number of regions. When this is no longer possible, if more than one region remained, each would give a nonempty open and closed subset of X, contradicting connectedness. ■

Combining Theorems 12.17, 12.15 and 12.9: every compact connected surface is homeomorphic to exactly one of S^2, T_n, P_m.